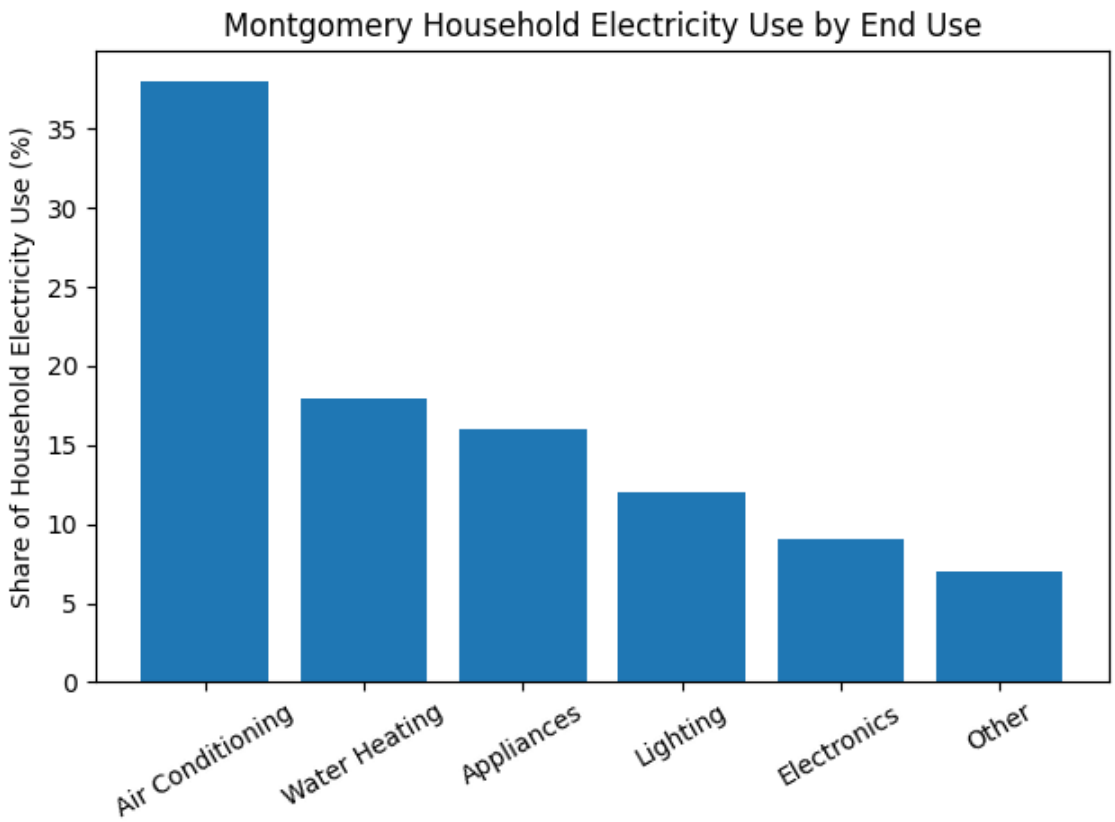
Montgomery Household Electricity Reduction Strategy 2030

Management Summary

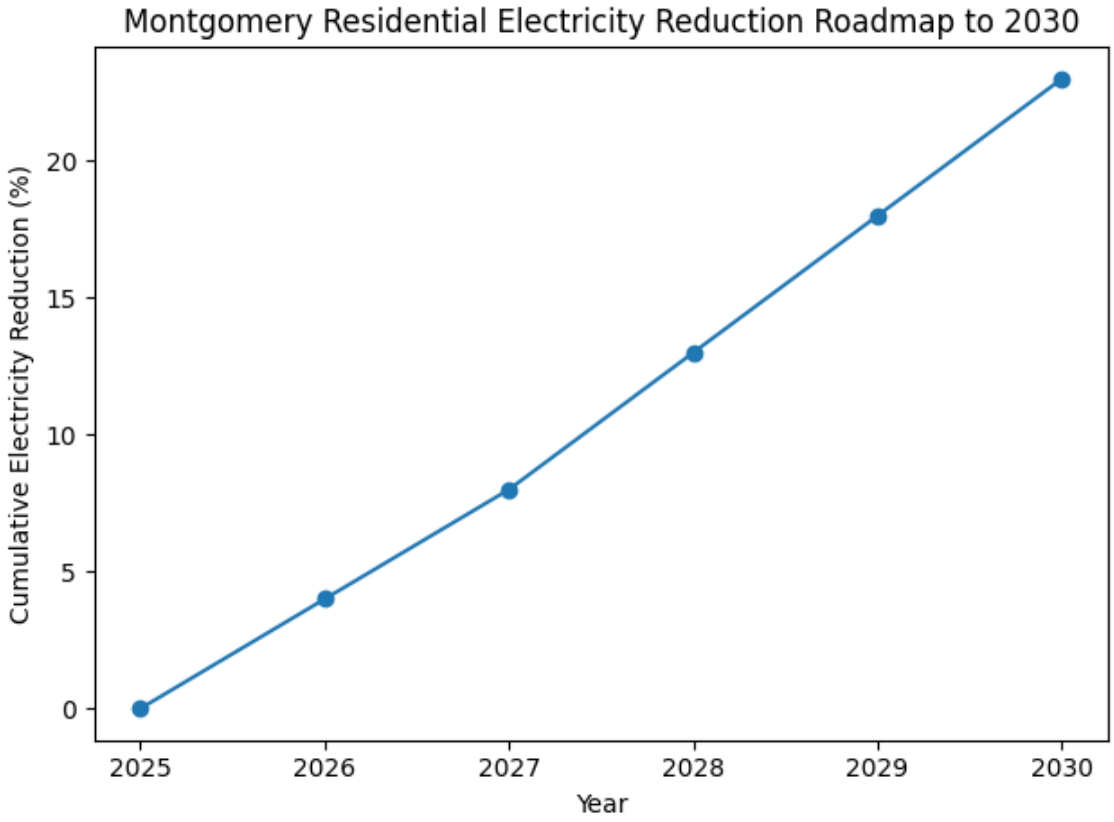
This report summarizes key insights on how Montgomery can reduce household electricity consumption by 20–25% by 2030. The strategy focuses on energy efficiency upgrades, smart technologies, and solar adoption. The charts below illustrate where electricity is used, how reductions can be achieved, and the financial benefits for residents and the city.

Household Electricity Use by End Use - Today



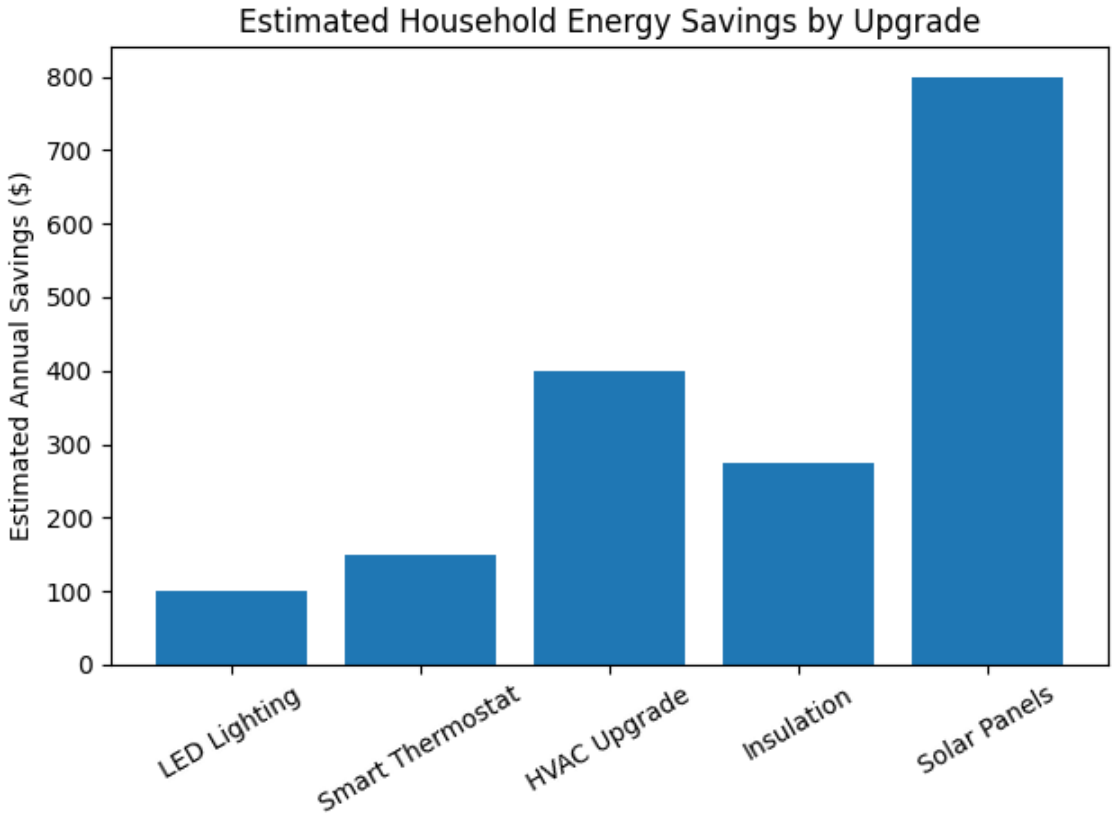
Cooling systems represent the largest share of household electricity consumption in Montgomery. Targeting air-conditioning efficiency and insulation improvements offers the largest reduction potential.

Residential Electricity Reduction Roadmap



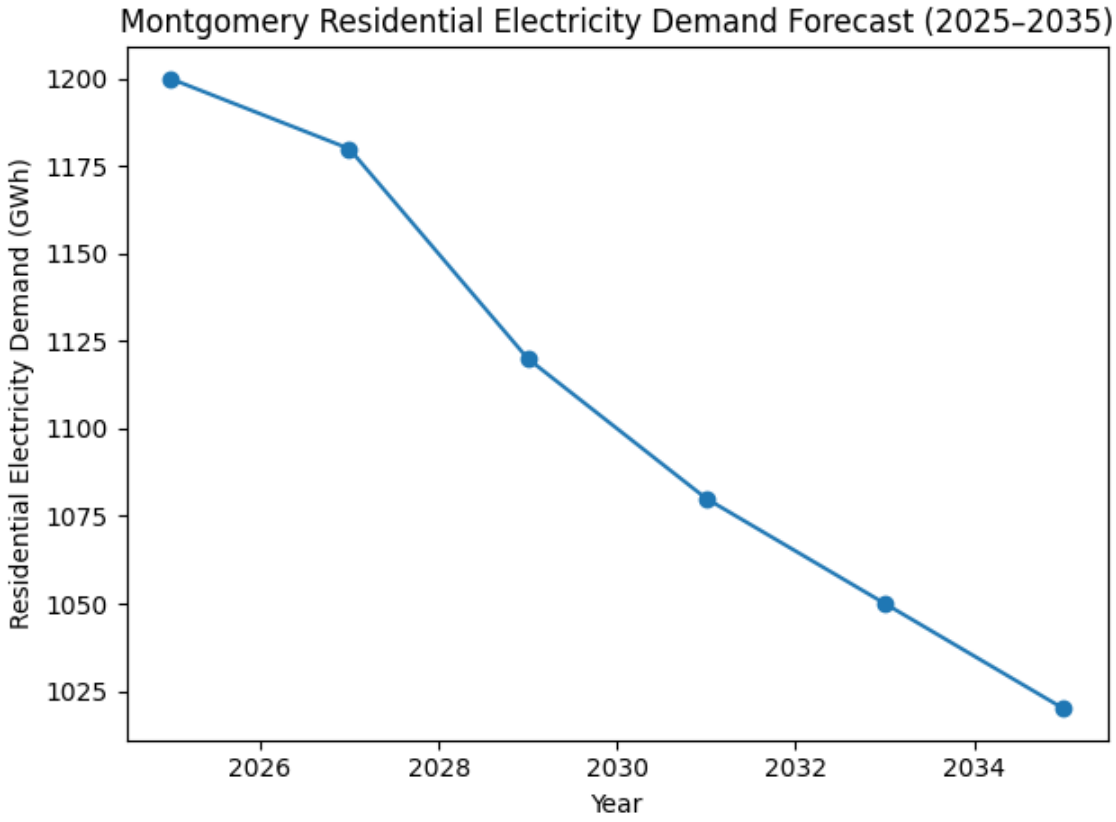
The roadmap illustrates how annual efficiency programs can gradually reduce residential electricity demand through 2030.

Estimated Household Savings by Upgrade



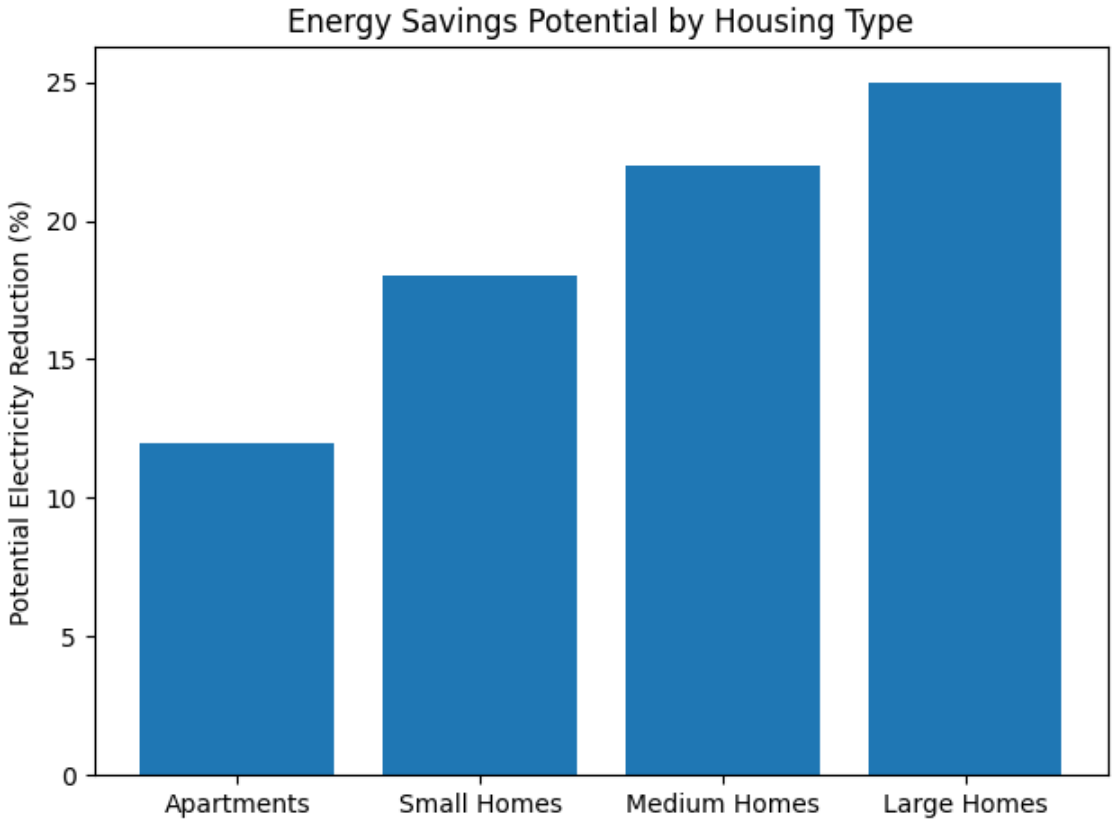
Energy efficiency upgrades deliver direct financial savings for residents. Solar installations and HVAC improvements provide the largest long-term savings.

Residential Electricity Demand Forecast



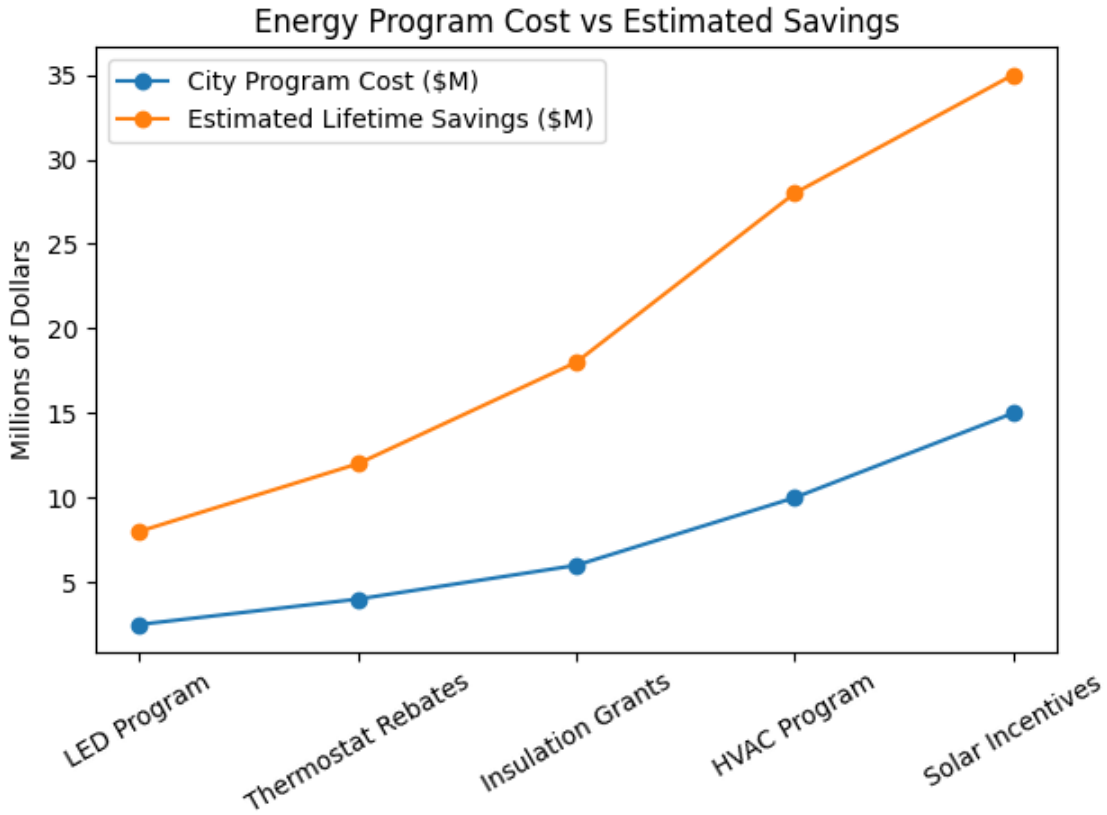
Forecast analysis shows that efficiency measures can reduce long-term electricity demand even as the city grows.

Energy Savings Potential by Housing Type



Larger homes offer the highest efficiency potential due to higher baseline electricity consumption.

City Energy Program Cost vs Savings



Investment in efficiency programs typically delivers multiple times the value in long-term electricity savings.

Example insight for City Council

Program	Cost	Estimated Savings	ROI
LED distribution	$2.5M	$8M	Very high
Thermostat rebates	$4M	$12M	High
Insulation grants	$6M	$18M	High
HVAC upgrades	$10M	$28M	Strong
Solar incentives	$15M	$35M	Strategic

Conclusion

Montgomery can significantly reduce residential electricity demand by **20–25% by 2030** through coordinated programs combining:

- household energy upgrades
- smart technology
- solar adoption
- public education
- municipal incentives

The strategy improves **affordability, grid reliability, and environmental sustainability**.